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Syd: A package for making interactive data visualizations in python

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Software

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Summary

Gaining an intuitive feel for the patterns in scientific data is fundamental to the scientific process. However, in the era of large datasets, it is not only challenging to understand the patterns in the data, but it is often a significant technical burden to simply plot the data in a satisfactory manner. For example, typical neuroscience datasets can have thousands to millions of neurons, spanning across recordings sessions and experimental subjects. Each neuron might contain a hint of a discovery. Similarly, studying the inner- workings of large-language models often comes down to looking at the activation patterns of thousands of neurons to understand how they represent text.

Syd represents a solution that supports interactive data analysis of rich datasets. It provides a python interface that enables the rapid construction of interactive plots, which enable scientists to quickly and easily look through all their data.

Statement of need

Although many packages enable the construction of graphical user interfaces in python, they all require some level of bespoke implementation and often depend on significant levels of boilerplate code. `Syd` relieves this need by providing a simple and opinionated interface for defining *what you want to plot* and *which parameters you want to be interactive*; it handles all the behind the scenes action required to make an interface.

`Syd` "viewers" are made by defining a plot function which accepts a `state` dictionary as input and returns a `matplotlib figure` object (Hunter, 2007). Then, parameters are added to the viewer with simple declarative functions. If required, `Syd` also provides an intuitive system for callbacks that greatly minimizes boilerplate GUI code. The simplicity of the interface means that interactive data visualization tools can become a fundamental part of a data analyst's typical workflow - as easy as making plots.

Because `Syd` handles the behind-the-scenes implementation code, it permits a user to "deploy" viewers in multiple environments without any changes to the code. Viewers can be deployed in jupyter notebooks for fast and local analysis or deployed on web browsers for sharing across local networks. These deployment environments depend on `ipywidgets` and `flask`, respectively (JupyterDevelopmentTeam, 2025; Pallets, 2024). In this way, a scientist can study their data at their desk, then use the same code to share the results interactively with their advisor or with their lab at different computers.

Example Viewer

This is a simple example that demonstrates how easy it is to make interactive visualization tools with Syd.

1. A plot function is defined that accepts `state` as input.
2. A viewer is created, and three parameters are added.
3. (Optional) Callback functions are defined and attached to parameters.
4. The viewer is deployed in the desired environment.

```python
import numpy as np
import matplotlib.pyplot as plt
from syd import make_viewer

# Plot function
def plot(state):
    amplitude = state["amplitude"]
    frequency = state["frequency"]
    color = state["color"]
    fig = plt.figure()
    t = np.linspace(0, 2 * np.pi, 1000)
    ax = plt.gca()
    ax.plot(t, amplitude * np.sin(frequency * t), color=color)
    return fig

# Viewer & Parameters
viewer = make_viewer(plot)
viewer.add_float("amplitude", value=1.0, min=0.1, max=5.0)
viewer.add_float("frequency", value=1.0, min=0.1, max=5.0)
viewer.add_selection("color", options=["black", "blue", "green", "red"])

# Deployment
viewer.show() # for viewing in a jupyter notebook
# viewer.share() # for viewing in a web browser
```

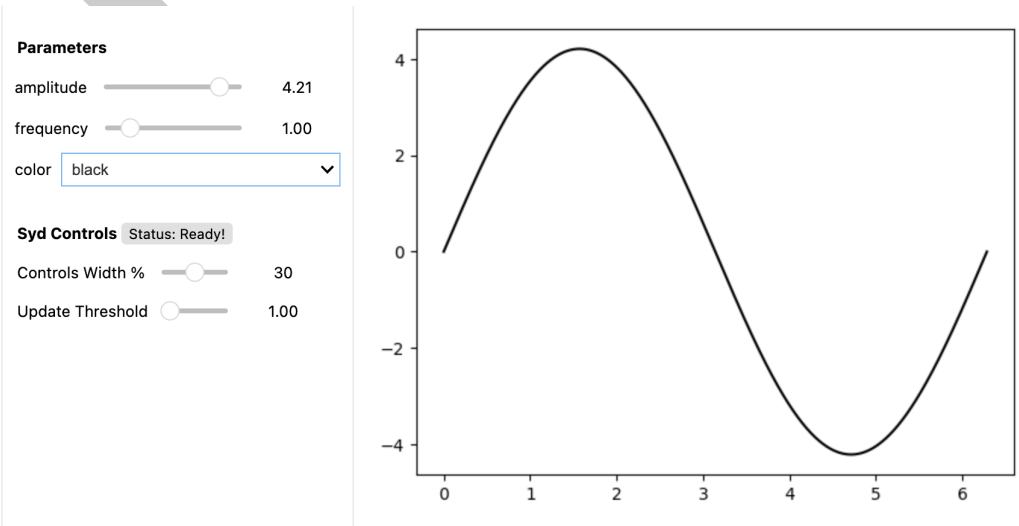

Figure 1: Example Syd Viewer

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